

Detailed Description

Parkinson Assistance project - full research groundwork. A 2-DOF finger exoskeleton (thumb + index) with EMG-triggered stiffening for tremor mitigation during grasping. Numbered citations [n] refer to the References section at the end.

Section 1 - Introduction and project definition

Parkinson's disease (PD) is the world's fastest-growing neurological disorder: 11.77 million people lived with PD in 2021, a 274% increase since 1990 [1], and modelling projects 25.2 million by 2050 [2]. Its cardinal motor signs are bradykinesia, rigidity and rest tremor [3]. Tremor of the hand is among the most visible and functionally disruptive symptoms, and a substantial share of patients get insufficient relief from medication or are not candidates for surgery. This creates a persistent need for assistive technology that acts directly on the hand.

Our project addresses this need with a wearable 2-DOF finger exoskeleton mounted around the hand, actuating the thumb and the index finger - the two digits that form the precision-grip couple. Surface electromyography (sEMG) sensors on the forearm and hand read the muscle activity driving these fingers. When the system detects the rhythmic 4-6 Hz muscle bursts characteristic of Parkinsonian tremor while the user is grasping, it stiffens the finger joints (raises the mechanical impedance) so the tremor is damped, while slow voluntary motion remains free. The device is therefore not a rehabilitation robot and not a grip amplifier: it is a tremor-contingent, grasp-contingent stiffening orthosis.

This document collects the research groundwork: the clinical background (section 2), existing treatments and their limits (section 3), the state of the art in tremor-suppression devices (section 4) and hand exoskeletons (section 5), control strategies (section 6), a deep dive on muscle signals for the fingers (section 7), design strategies for our device (section 8), and an evaluation methodology (section 9).

Section 2 - Parkinson's disease background

2.1 Motor symptoms relevant to hand function

Parkinsonism is clinically defined as bradykinesia (slowness with progressive decrement of repetitive movement) combined with rest tremor and/or rigidity [3]. The classic Parkinsonian tremor is a rest tremor at 4-6 Hz (reported band 3.5-7.5 Hz), typically asymmetric and often described as pill-rolling because the thumb and index finger oscillate against each other [4]. Amplitude is largest distally: rest tremor displacement is most evident at the fingers, which is why clinical accelerometry is standardly performed at the index finger with band power measured at 4-7 Hz [4]. Limb rest tremor affects about 65-70% of PD patients, and patient organizations report that up to 75% (70-90% across sources) experience tremor during the disease [5, 6].

2.2 Re-emergent tremor - the key phenomenon for grasping

Rest tremor classically attenuates when a movement starts. However, when the arm adopts a sustained posture - holding a cup, a spoon, a phone - the tremor reappears after a latency of roughly 10 seconds (reported mean 9.4 +/- 10.7 s, later studies about 16 s, range 1 to over 60 s) at the same 4-6 Hz frequency. This is called re-emergent tremor [5]. It responds somewhat less to levodopa than pure rest tremor. The consequence for our project is central: static holding is not protected by the movement-related suppression of tremor, so a device that stiffens the fingers during grasp-and-hold addresses exactly the phase of daily activities where Parkinsonian tremor strikes.

2.3 Functional impact

In a 2025 survey of 634 people with PD tremor [6]: 65% reported at least mild functional impairment from tremor; 83% said at least one daily activity was affected (handwriting 60%, keyboard use 52%, smartphone use 47%, work ability 33%, sleep 30%); and 92% reported psychosocial burden such as self-consciousness and hiding the hand. Finger-level tremor amplitude scales with clinical severity: one accelerometry study reports average movement paths of about 11.5 mm at UPDRS tremor score 1, 33.6 mm at score 2, and 382 mm at score 3 - millimeters to centimeters of oscillation at the fingertip [7].

2.4 Parkinsonian tremor versus essential tremor

Essential tremor (ET) is the other major tremor disorder and the target of most commercial devices, so the distinction matters. PD tremor: 3-6 Hz, occurs at rest, suppressed at movement onset, re-emerges during posture, asymmetric. ET: 4-12 Hz (commonly 4-8), action/postural tremor appearing within 1-2 seconds of adopting a posture, usually bilateral, often involves head and voice [8]. In the EMG, PD tremor bursts typically alternate between antagonist muscles while ET bursts are synchronous - a diagnostic feature we can exploit (section 7.5). Devices validated only in ET (action tremor at the wrist) do not automatically transfer to PD grasping scenarios: a PD-targeted device must handle 4-6 Hz rest/re-emergent tremor that is strongest at the fingers.

Section 3 - Current interventions and their limitations

Levodopa remains the foundation of PD therapy, and tremor does respond: in advanced PD, proportional improvement of tremor (87%) exceeded bradykinesia (46%) and rigidity (67%), with complete tremor resolution in 68% of patients on high-dose therapy [9]. But at ordinary clinical dosing, about 39% of patients have rest tremor resistant even to dispersible levodopa challenges - a dopamine-resistant tremor cluster attributed to cerebello-thalamic circuits that are less dopamine-sensitive [10]. Chronic levodopa therapy also causes motor fluctuations and dyskinesia in up to about half of patients within 2-5 years and 80-100% by 10 years, with tremor re-emerging between doses during wearing-off [11].

Deep brain stimulation (DBS) is the most effective treatment for tremor: thalamic (VIM) DBS gives around 70% tremor improvement at 1 year and 63% beyond 10 years, and STN-DBS additionally treats bradykinesia and fluctuations [12]. But it is brain surgery: device cost around 21,500 USD and total surgical episode around 40,900 USD on average [13], roughly 2% risk of intracerebral hemorrhage, 2-3% infection, and eligibility criteria (no dementia, good levodopa response, acceptable comorbidity) that exclude the majority of patients.

MRI-guided focused ultrasound (MRgFUS) thalamotomy is an incisionless alternative: in the pivotal randomized trial for tremor-dominant PD, on-medication hand tremor scores improved a median 62% versus 22% for sham [14]. It is, however, a unilateral, irreversible lesion with its own eligibility constraints.

The bottom line motivating assistive devices: a large residual population - dopamine-resistant tremor in roughly 4 of 10 patients, fluctuating benefit in many more, and invasive options that most never receive - lives with tremor that disrupts grasping. Wearable mechanical or electrical tremor suppression targets precisely this group [15, 16].

Section 4 - State of the art: tremor-suppression devices

4.1 Strategy taxonomy

The literature sorts suppression approaches into five families [15, 17]: (a) passive mechanical loading and damping (constant added mass, viscosity or friction - simple, but it loads voluntary motion too); (b) active counteraction (motors inject torque opposing the tremor at its frequency - most authority, but heavy, power-hungry, and capable of injecting energy if the controller errs); (c) semi-active modulation (elements that only dissipate or lock - magnetorheological (MR) fluid dampers, brakes, jamming structures - inherently unable to amplify tremor); (d) functional electrical stimulation (FES) of muscles, either out-of-phase with the tremorogenic bursts or as co-contraction stiffening; and (e) peripheral afferent stimulation below motor threshold, which disrupts the central tremor network through sensory pathways.

4.2 The canonical research device: WOTAS

WOTAS (Wearable Orthosis for Tremor Assessment and Suppression) [18] is the reference active tremor exoskeleton: 3 DOF (elbow flexion/extension, forearm pronation/supination, wrist flexion/extension), DC motors, 850 g. It implemented both an impedance mode (adding virtual viscosity and inertia to the tremor component only) and an active cancellation mode with notch filtering at the tremor frequency. Across 10 patients with ET, PD and MS it achieved on average about 40% tremor power reduction and up to 80% in severe tremor; mode-resolved figures are about 70% mean power reduction in impedance mode and 81% in active mode, with effectiveness dropping for very low tremor power because forces leak through soft tissue [18, 19]. WOTAS also flagged two lasting lessons: weight and bulk drive rejection by users, and in one patient suppression at the wrist was associated with a possible increase of shoulder tremor - so proximal tremor migration must be monitored [19].

4.3 Commercial and near-commercial devices

GyroGear GyroGlove: a control-moment gyroscope on the back of the hand resists angular deviation - passive-reactive stabilization with no tremor-sensing loop. Launched commercially in 2024 at 5,899 USD, FDA Class I registered; total mass around 580 g (stabilizer plus battery). No completed controlled trial has been published; manufacturer figures (about 50% typical suppression) are unverified [20]. Steadiwear Steadi-Two/Steadi-3: battery-free glove stabilizing the wrist with a tuned-mass damper plus a smart-fluid ball joint that stiffens under tremor-rate motion. The predecessor showed 85-90% suppression in laboratory simulation only; no published clinical trial data [20, 17].

Liftware Steady: active-cancellation cutlery - accelerometers sense tremor and voice-coil motors counter-move the spoon; validated in ET with 71-76% reduction in utensil tremor [21]. Task-specific (eating only), but proof that active cancellation works in a palm-sized package. Cala Trio/klQ: wrist-worn transcutaneous afferent patterned stimulation (TAPS) alternating median and radial nerve stimulation at the individually calibrated tremor frequency. In an open-label PD action-tremor trial (n=40): median 64% tremor-power improvement, 79% of patients above 50% reduction [22]. Effects are session-bound. Other devices include the Tremelo passive tuned-mass absorber, the Readi-Steady weighted glove, and the TAPO task-adjustable passive air-filled orthosis (74-82% reduction across three daily activities at only 33 g) [17].

4.4 FES and stimulation results

Out-of-phase FES of antagonists reported 62-73% tremor reduction in early work (Javidan/Prochazka: ET 73 +/- 14%, PD 62 +/- 5%) and 50-67% in modern implementations; co-contraction stimulation (electrically stiffening the joint - conceptually identical to our mechanical stiffening) achieved 52 +/- 26% (Gallego et al., wristband neuroprosthesis) [16, 23]. Limitations are muscle fatigue, discomfort and electrode sensitivity. Sub-motor-threshold sensory stimulation avoids fatigue but is highly variable (12-90% across studies) [16].

4.5 Aggregate numbers and the recurring gaps

Across 25+ wearable orthoses, a 2021 review reports pooled efficacy around 83 +/- 13%, with bench results (95 +/- 5%) systematically higher than human-subject results (70-86%); active devices average 561 g versus 191 g for passive ones [15]. Two structural gaps recur. First, sensing: 92% of devices rely on motion sensing (IMU) alone - only one reviewed device used EMG [15]. Second, location: nearly all devices act at the wrist, forearm or elbow, or stabilize the utensil; the fingers themselves are almost untouched (section 5.5). None of the reviewed research orthoses reached the market, and several commercial devices lack published trials [15, 20].

Section 5 - State of the art: finger and hand exoskeletons

5.1 What exists and what they are used for

Hand exoskeletons cluster in four application domains [24, 25]: stroke/SCI rehabilitation (repetitive therapy - Gloreha, Hand of Hope), assistive grasping for weak grip in daily life (Bioservo Carbonhand/SEM Glove, Neofect NeoMano, HERO glove, RELab Tenoexo), haptics/VR and teleoperation (CyberGrasp, Dexmo, HaptX, Festo ExoHand), and research platforms. Representative specifications: Carbonhand - soft glove, remote DC motors pulling tendons to three fingers, up to 20 N per finger, commercial, used as a grip amplifier triggered by force sensors; Harvard/Polygerinos soft glove - fiber-reinforced elastomeric actuators, hydraulic waist pack, about 285 g glove, home rehabilitation [26]; Exo-Glove (SNU) - soft tendon-routing glove for thumb+index+middle, 20 N pinch, 194 g glove mass [27]; Hand of Hope - rigid dorsal linkages, one linear actuator per finger driving coupled MCP+PIP flexion, EMG-triggered stroke therapy [24]; RELab Tenoexo - three-layered sliding-spring mechanism, hand module under 150 g with 720 g back module, 4.5 N per finger covering about 80% of grasp types [28]; HERO grip glove - 80 N linear actuators on a wrist brace,

tendons for flexion and extension, stroke assistance [29]. Thumb+index devices form a recognized sub-genre (series-elastic thumb exoskeletons, six-bar pinch mechanisms, EMG-driven soft pinch exoskeletons), validating thumb+index as the minimal functional grasp set [24].

5.2 Mechanism taxonomy

Four structural families dominate [24, 25]: rigid linkages with matched joint axes (simple but misalignment causes parasitic loads); rigid linkages with remote centers of rotation (the rotation axis is synthesized inside the finger joint by a four-bar or circular guide - avoids lateral bulk and tolerates finger-size variation); tendon/Bowden-cable driven gloves (actuators moved off the hand to the forearm, waist or wheelchair - minimal hand mass, but cable friction and hysteresis); and soft pneumatic/fluidic bending actuators along the fingers (inherently compliant and safe, force distributed along the finger, but tethered to pumps and of limited stiffness). Underactuation is the dominant trend: one actuator drives multiple joints or fingers through differentials or compliant elements, giving adaptive grasping without per-joint motors.

DOF choices: anatomically a finger has 4 DOF and the thumb about 5, but deployed designs use 1-2 DOF per finger, because PIP and DIP joints are anatomically coupled, grasping (as opposed to dexterity) is achievable with coupled flexion, and every added actuator costs mass on the hand. The thumb is usually handled by locking or passively supporting the CMC joint in opposition and actuating only thumb flexion. Our 2-DOF choice (one coupled flexion DOF per digit) is therefore mainstream practice, not a compromise [24].

5.3 Actuation technologies and their trade-offs

Motion-producing actuators: DC motors with cable/Bowden transmission dominate (remote placement keeps hand mass low); linear actuators with self-locking gearboxes hold force without power (Hand of Hope, HERO, and the 2025 HELM tremor exoskeleton use exactly this); pneumatics give high force-to-weight but need pumps and valves; series elastic actuators (SEA) put a spring between motor and joint, turning position control into accurate torque control with inherent compliance; shape memory alloys are extremely light (a 15 g glove produced 15 N) but their cooling-limited bandwidth (seconds per cycle) rules them out for the 4-6 Hz tremor band [24, 25].

Stiffness/damping-modulating (semi-active) elements - the most relevant class for us: MR-fluid brakes respond in milliseconds (5-15 ms force changes) with miniature rotary designs reaching about 0.9 Nm braking torque at 85 g or 0.3 Nm at 122 g; they dissipate only, draw little power, and have already been used in wrist tremor suppressors (WTSE: 262 g, 8 N damping force; SETS soft exoskeleton: 61.9% acceleration reduction) [30, 31]. Layer jamming (vacuum-stiffened laminates) is featherweight: a 30 g layer-jamming tremor glove achieved 74.8-78.3% amplitude reduction on a tremor simulator and 24.7-42% band-power reduction in an 11-patient clinical test - but requires a vacuum source and switches state more slowly [32]. Electrostatic/electroadhesive clutches hold up to 100 N per 1.5 g electrode pair, switch in under 30 ms, and need almost no holding power - attractive as tendon locks, though binary rather than proportional [33]. Friction brakes and self-locking lead screws give hold-without-power. Antagonistic tendon co-contraction can render variable stiffness but costs two actuators and continuous power.

5.4 How existing hand exoskeletons are controlled

Reported schemes range from passive trajectory playback (Gloreha) and button/app control (NeoMano) through force-sensor triggered amplification (Carbonhand FSRs detect the user's own grip and amplify proportionally) to EMG: threshold-triggered opening/closing (Hand of Hope), gross-activation detection for assistive gloves (Polygerinos), EMG pattern recognition of hand motions, and EMG-proportional control [24, 26]. EMG-triggered grasp assistance is thus proven technology - our novelty is EMG-triggered tremor stiffening, not EMG control per se.

5.5 Tremor-targeting finger/hand exoskeletons: the gap

Dedicated searching confirms finger-level tremor suppression is exceptionally rare. The complete inventory found: the Wearable Tremor Suppression Glove (WTSG, Western University, IEEE TBME 2021) - DC motors and cables suppressing tremor independently at index MCP, thumb MCP and wrist; suppression 73-86% (resting), 70-81% (postural), 59-65% (kinetic) across joints, tested with recorded Parkinsonian tremor; motivated explicitly by the team's finding that suppressing at elbow/wrist can exaggerate finger tremor [34]. HELM (2025) - five light linear motors, 213 g total, over 80-90% reduction with simulated tremor and over 70% in one PD patient; its authors state it is the first five-finger tremor-suppression hand exoskeleton - published in 2025, showing how sparse this field still is [35].

The 30 g layer-jamming glove (section 5.3) stiffens continuously, not on demand [32]. A pneumatic-muscle soft glove attenuated index-finger tremor by about 75% in a closed loop [36]. Commercial devices (GyroGlove, Steadiwear, Cala, Liftware) all stop at the wrist or the utensil.

Gap statement: no found device combines (i) finger-level action at thumb + index, (ii) semi-active stiffening rather than motorized counter-motion, and (iii) EMG-based detection making the stiffening tremor- and grasp-contingent. The WTSG is motor-based and IMU-driven; the jamming glove is stiffness-based but always on; EMG-in-the-loop closed-loop suppression exists only in FES form (section 6.7). This combination is a defensible novelty for the project.

Section 6 - Control of exoskeletons and tremor-suppression algorithms

6.1 Control architectures

Position control commands a trajectory regardless of interaction forces - it fights any voluntary motion and suits only rigid guidance. Force/torque control regulates interaction torque directly and is the usual inner loop. Impedance control (Hogan) makes the device render a virtual spring-damper: $\text{torque} = K \times (\text{position error}) + B \times (\text{velocity error})$ about a reference; it degrades gracefully and is preferred on back-drivable hardware. Admittance control is the dual (sense force, command motion) used on stiff, geared systems, but it is documented to become unstable when the human limb stiffens against it - a direct concern when meeting rigid, tremulous co-contraction [37]. For our device the literature points to impedance control about the estimated voluntary trajectory: near-zero rendered impedance below about 2 Hz, high stiffness/damping against the 4-6 Hz band - exactly the WOTAS impedance mode [18].

Two properties matter for wearables: transparency (no resistance during free voluntary motion, quantified as interaction torque during passive movement) and back-drivability (torque needed to move the unpowered device). Semi-active elements are transparent when off by construction - a core argument for brakes/jamming over motors.

6.2 Rendering variable stiffness

The control-theoretic frame: estimate the voluntary trajectory, then render torque = $K \times (\text{voluntary estimate} - \text{actual position}) + B \times (\text{velocity of estimate} - \text{actual velocity})$. The impedance (K , B) is anchored to the moving voluntary set-point, so stiffness and damping act only on the tremor deviation. Implementation options map to hardware: software impedance on a motor (needs good torque control and bandwidth well above 6 Hz), series elastic actuators (the spring doubles as torque sensor; note small-scale SEAs can have force-control bandwidth as low as about 2.5 Hz - marginal against 4-6 Hz tremor), variable stiffness actuators (a second motor changes spring preload - heavier), or semi-active dampers/brakes/jamming that modulate only dissipation (unconditionally stable, cannot inject energy) [37, 30].

6.3 The signal-separation problem

Voluntary movement in daily activities occupies roughly 0-2 Hz (predominant component near 1 Hz, 75% of spectral energy below 5 Hz) [38]; Parkinsonian tremor sits at 4-6 Hz, other pathological tremors 4-12 Hz [4]. The guard band between 2 and 4 Hz is narrow, so naive sharp filtering introduces phase lag that both distorts the voluntary estimate and mis-phases any active cancellation. The standard real-time estimators are: the Fourier Linear Combiner (FLC - adaptive weights on a sine/cosine pair at fixed frequency, zero phase lag); the weighted-frequency FLC (WFLC - also adapts the frequency; reduced RMS tremor motion by 67% in its original application) [39]; the band-limited multiple FLC (BMFLC - a bank of FLC dyads spanning 3-9 Hz; 64% cancellation versus 43% for WFLC in comparative tests) [40]; adaptive notch filters (the WOTAS active mode); and Kalman-filter schemes, including two-stage estimators where a g-h (Benedict-Bordner) filter tracks the voluntary motion and the residual feeds a WFLC - effective delay about 1 ms [19, 41]. All are per-sample recursions of a few multiply-accumulates: comfortably microcontroller-class at 100-500 Hz loop rates.

6.4 Tremor detection for triggering stiffening

The standard trigger is a threshold on band power (3-12 Hz) from an IMU or the EMG envelope. Spectral methods (FFT/STFT) need windows of 1-3 s for adequate frequency resolution, implying detection latency of the same order; adaptive-filter residual tracking avoids that trade-off. An optimized second-order-moment detector on sEMG localized tremor burst onsets/offsets with bias and standard deviation under 8 ms [42]. Machine-learning detectors on wrist wearables report sensitivity around 0.90 for free-living PD tremor [43]. Crucially, EMG precedes the mechanical oscillation by the electromechanical delay of roughly 20-100 ms (section 7.6), so an EMG-based trigger can engage stiffening before the shake grows - the single strongest argument for EMG sensing in this device [44].

6.5 Intention detection and shared control

The canonical dichotomy in assistive devices is EMG-triggered (binary onset detection arms or fires a behavior) versus EMG-proportional (continuous mapping of EMG amplitude to assist level) [45]. Devices avoid fighting the user by anchoring impedance to the voluntary estimate (section 6.2) or by variable admittance whose damping drops when voluntary sEMG is detected. For our device the natural scheme is: EMG grasp detection (flexor co-activation above threshold) arms the system; EMG/IMU tremor detection within the grasp state engages stiffening; stiffness scales with detected tremor power (a semi-proportional scheme).

6.6 Safety

Published finger exoskeletons cap driving torque near 0.1 Nm per joint with current limits plus hard mechanical end-stops at anatomical joint limits [46]; finger joint torques in daily grasping span about 0.06-1.4 Nm, so the device must resist tremor (small oscillating torques) without ever overpowering the user. Semi-active stiffening is inherently the safest class: it can only oppose motion, never drive it. Remaining risks to manage: admittance-style instability against stiffened joints, mis-tuned active cancellation amplifying off-frequency motion (excluded by choosing semi-active), discomfort from load concentration (distribute pressure, balance torque across MCP/PIP), and proximal tremor migration - monitor wrist/elbow/shoulder tremor in evaluation, since one WOTAS patient showed a possible shoulder increase, though the majority showed no distal-to-proximal shift [19].

6.7 EMG-in-the-loop precedents

Closed-loop EMG-driven tremor suppression has been demonstrated only with electrical stimulation: Dosen et al. (2015) drove out-of-phase sensory-level stimulation from antagonist EMG demodulated with an iterated Hilbert transform (42% acute reduction); Gallego et al. (2013) characterized tremor in real time and countered it with co-contraction stimulation (52 +/- 26%); Widjaja et al. (2011) exploited the electromechanical delay for anti-phase FES [23, 44, 47]. Mechanical devices remain IMU-driven (WTSG included). A mechanical exoskeleton with EMG closing the suppression loop appears to be a genuine open niche.

Section 7 - Muscle signals to the fingers: the deep dive

7.1 Where the signal comes from

The functional unit of muscle control is the motor unit: one alpha motoneuron plus all muscle fibers it innervates. Each discharge produces a motor unit action potential (MUAP); force is graded by recruiting more units (small to large, Henneman's size principle) and by rate coding (firing rates from about 8-10 pulses per second at recruitment toward 35-50 near maximal effort in hand muscles) [48]. A surface electrode sees the spatiotemporal summation of tens of concurrent MUAP trains, each filtered by the tissue between fiber and skin. The practical numbers: amplitude 0-10 mV peak-to-peak (0-1.5 mV RMS), usable energy 0-500 Hz with dominant energy 50-150 Hz; the recording band is set to about 20-450 Hz because below 20 Hz movement artifact dominates and above 450 Hz there is little EMG energy left [49].

7.2 Which muscles drive the thumb and index finger

Extrinsic muscles (bellies in the forearm, tendons into the hand): flexor digitorum superficialis (FDS - PIP flexion of digits 2-5, mid-anterior forearm, reasonably surface-accessible); flexor digitorum profundus (FDP - DIP flexion, lies deep under FDS, poor selective surface access); flexor pollicis longus (FPL - thumb flexion, deep radial-anterior forearm, partly estimable from the surface); extensor digitorum communis (EDC - extension of digits 2-5, superficial on the dorsal forearm, the best surface target for finger extension, with partially separate per-finger bellies); extensor indicis (independent index extension but small, deep and distal - not a realistic surface target); extensor pollicis longus/brevis and abductor pollicis longus (dorso-radial distal forearm, small and overlapping) [50, 51].

Intrinsic hand muscles (superficial and highly selective - the clinical favorites): first dorsal interosseous (FDI - index abduction and MCP flexion assist, extremely superficial in the dorsal first web space) and abductor pollicis brevis (APB, thenar eminence - thumb abduction/opposition). Both are standard sites in PD tremor neurophysiology precisely because they are easy to isolate [51].

Recommended montage for our device: FDS and EDC on the forearm for index flexion/extension drive, plus FDI and thenar/APB on the hand for digit-specific and tremor-sensitive channels - four channels minimum, six to eight preferred (section 7.4).

7.3 Acquisition practice

Electrodes: pre-gelled Ag/AgCl, conductive diameter 10 mm or less, bipolar pairs with 20 mm center-to-center spacing aligned with the fiber direction, placed mid-belly between the innervation zone and the distal tendon - never on the innervation zone (SENIAM guidance); reference electrode on electrically inactive tissue (wrist bone, olecranon, C7) [52]. For the small, crowded forearm muscles a smaller inter-electrode distance (about 10 mm) measurably reduces crosstalk [53]. Skin prep: shave, light abrasion, alcohol. Dry electrodes and armbands trade fidelity for convenience.

Electronics: bipolar differential amplification with CMRR above 80 dB (research amplifiers 90-110 dB), input impedance 100 megohm or more, gain 500-2000. Sampling at 1 kHz minimum (Nyquist for the 450-500 Hz content); 2 kHz is common in research; 12 bit or better. Hardware classes: research-grade (Delsys Trigno: 20-450 Hz bandwidth, under 0.75 microvolt RMS noise; OT Bioelettronica HD grids), embedded/student-grade (MyoWare 2.0: onboard band-pass 20.8-498 Hz and envelope output - digitize its RAW output at 1 kHz or more and do the processing yourself; BITalino: 25-482 Hz, 1 kHz, 10 bit), and armbands (gForce 8-channel; the discontinued Myo sampled only 200 Hz - a warning that undersampled hardware cannot support proper spectral work) [54].

7.4 The processing pipeline (sorting the signal)

Step 1 - band-pass 20-450 Hz (removes motion artifact below 20 Hz and out-of-band noise). Step 2 - powerline handling: in Europe interference sits at 50 Hz, inside the EMG's dominant band; classic advice is to fix it at the source (electrode contact, short leads, battery power) rather than notch-filter, and if unavoidable use a narrow adaptive notch [49]. Step 3 - rectify (full-wave) and extract the envelope: a moving RMS or mean-absolute-value window. Here our application forces a key choice: for grasp-intent detection the myocontrol literature recommends 150-250 ms windows (smooth, stable, total controller delay under 300 ms) [55], but a 150-250 ms window smears 4-6 Hz bursts away - for tremor visibility the envelope

bandwidth must exceed the tremor frequency, so use a fast envelope (low-pass around 8-10 Hz, or 50 ms windows or shorter) in parallel with the slow intent envelope. Two envelopes from the same channels serve the two jobs.

Step 4 - normalize amplitudes to a reference contraction (percent of maximum voluntary contraction, MVC); for PD users a standardized submaximal reference contraction may be more repeatable than a true MVC [56]. Step 5 - onset/burst detection: single threshold (baseline mean plus 2-3 standard deviations), double-threshold detectors, or Teager-Kaiser Energy Operator (TKEO) preconditioning, which cut onset-timing errors from 229 +/- 356 ms to 40 +/- 99 ms in comparative tests [57]. Step 6 - features and classification where needed: the Hudgins time-domain set (MAV, zero crossings, slope-sign changes, waveform length) plus RMS; mean/median frequency for fatigue tracking; LDA or SVM classifiers reach over 90% accuracy on individuated finger movements with multi-channel forearm recordings (12 classes, 19-32 electrodes; about 98% for 15 classes with an optimally chosen 6 channels) - but with only 2 channels robust discrimination collapses to a few gross classes, hence the 4-8 channel recommendation [58, 59].

7.5 What tremor looks like in the EMG

Parkinsonian rest tremor produces rhythmic grouped EMG bursts at 4-6 Hz with a narrow spectral peak and harmonics; the hallmark is alternating bursting between antagonist muscles (flexors versus extensors), whereas essential tremor bursts are usually synchronous in antagonists [60]. Quantified: alternating patterns in 15 of 15 PD rest tremors versus synchronous in 15 of 15 ET cases in one study [61]; alternating contraction in 80% of PD versus synchronous in 80% of ET in a 92-patient series, with PD rest tremor at 4.52 +/- 0.43 Hz [62]; a 2024 97-patient series reports about 25% atypical synchronous PD cases - the pattern is typical, not universal.

Standard quantification: rectify and low-pass the EMG, compute the envelope power spectrum (tremor appears as a sharp 4-6 Hz peak), track band power in 3-9 Hz, and use EMG-accelerometer coherence at the tremor frequency. The tremor stability index (cycle-to-cycle frequency jitter) separates PD from ET with about 90% accuracy [63]. Beyond detection, the iterated Hilbert transform demodulates tremor amplitude, frequency and phase plus the degree of voluntary activity directly from sEMG [64], and hidden-Markov approaches have predicted upcoming tremor episodes from EMG before they appear mechanically [65].

7.6 The electromechanical delay: our prediction window

Because muscle force develops after electrical activation, sEMG precedes the measured tremor motion by roughly 20-100 ms [44]. This window is what makes EMG the right trigger for a stiffening device: detect the tremorogenic burst pattern electrically, engage stiffening before the mechanical oscillation peaks. This is exactly the mechanism exploited by EMG-driven out-of-phase FES systems (section 6.7), transplanted to a mechanical actuator.

7.7 Separating tremor from voluntary drive in the EMG

The voluntary drive lives in the low envelope band (predominantly below 2-3 Hz for daily-activity motion) while the tremor band is 3.5-7.5 Hz for PD; band-energy comparisons (for example 0.1-3 Hz versus 4-12 Hz envelope power) form simple robust classifiers, and the WFLC/BMFLC adaptive filters (section 6.3) apply equally to EMG envelopes, tracking the tremor component online with zero phase lag [39, 40]. In practice the discriminator is the

combination of rhythmicity (narrow spectral peak, stable cycle timing), band location (4-6 Hz), and the alternating antagonist pattern - voluntary grasp adjustments are broadband, non-rhythmic and co-contracting.

7.8 PD-specific confounds and practical pitfalls

Confounds to design for: rigidity raises tonic background EMG, so envelope thresholds drift with the patient's state (levodopa measurably reduces the involuntary tonic activity) [66]; medication ON/OFF changes tremor expression - detectors must be validated in both states; tremor amplitude is far more labile than frequency (stress and mental load increased PD tremor in 100% of one cohort), and frequency shifts slightly between rest and posture (about 4.5 to 4.9 Hz) - so track the tremor peak adaptively rather than hard-coding 5 Hz; muscle fatigue compresses the EMG spectrum toward lower frequencies over long sessions [62, 66].

Student-project pitfalls: electrode shift of even 1 cm between donnings degrades classifiers - mark positions with a skin pen or a template and recalibrate every session [67]; motion artifact and cable sway live below 20 Hz - use the high-pass and secure short leads; sweat and drying gel drift the baseline; crosstalk between neighboring forearm compartments is unavoidable - mitigate with smaller electrode spacing, mid-belly placement away from muscle borders, and decision logic that uses multiple channels; avoid the tendon-crowded wrist region; battery-power the front end while debugging 50 Hz issues; and budget a short per-session calibration routine (rest baseline, reference contraction, 30-60 s tremor sample to lock the personal tremor frequency).

Section 8 - Design strategies for our device

8.1 When to stiffen (control policy options)

Option A - always-on damping: simplest, proven by the layer-jamming glove (25-42% clinical reduction), but permanently loads voluntary motion and fatigues the user. Option B - grasp-triggered stiffening: EMG flexor co-activation arms the device; stiffening engages during grasp-and-hold. Catches the re-emergent tremor scenario; simple state machine. Option C - tremor-triggered stiffening (our recommendation): grasp detection arms; tremor detection (4-6 Hz band power over threshold in the fast EMG envelope, confirmable by IMU) engages; stiffness scales with tremor power; release when tremor subsides or the grasp opens. This is the tremor- and grasp-contingent policy no existing device implements. Option D - full active cancellation: inject anti-phase torque (WOTAS active mode, WTSG) - highest suppression ceiling but requires motors with bandwidth well above 6 Hz, adds mass, and can inject energy; not recommended for a first prototype.

8.2 How to stiffen (actuation options ranked)

For a 2-DOF thumb+index device with a hand-mass budget under 200-300 g: (1) miniature MR brake or damper per digit - proportional damping, millisecond response, 0.3-0.9 Nm class at 85-122 g per unit - the strongest match to the requirements if the mass fits; (2) layer/granular jamming elements along the fingers - featherweight (30 g demonstrated), large stiffness ratio, needs a small vacuum pump (can live on the forearm or belt) and switches slower; (3) electroadhesive tendon clutch - grams-scale, under 30 ms switching,

binary lock of a tendon across the joints (stiffen = lock the tendon length); (4) small DC motor with tendon and self-locking gearing - doubles as an assistive actuator if desired later, but adds control complexity. A pragmatic architecture: passive glove with dorsal leaf springs for finger extension bias, one stiffening element per digit acting on a flexion tendon, electronics and any pump/battery on the forearm.

8.3 Sensing architecture

EMG: 4 channels minimum - FDS, EDC (forearm), FDI, thenar/APB (hand) - sampled at 1 kHz or more, 12 bit or better; two envelope paths per channel (slow 150-250 ms for intent, fast 8-10 Hz for tremor bursts). IMU: one 6-axis unit on the index proximal phalanx or hand dorsum as ground truth for tremor (4-7 Hz band power) and as a second trigger vote; IMU-based detection also covers sessions where EMG quality degrades. Optional: a force-sensitive resistor at the fingertip pulp for grasp confirmation - proven in the Carbonhand. Fusion policy: grasp state from the slow EMG envelope (plus FSR if present); tremor state from fast EMG envelope band power and/or IMU band power; stiffen only when grasping and tremor coincide.

8.4 Controller

A layered scheme, all microcontroller-class. Layer 1 (state machine): idle, then grasp-armed, then stiffened - with hysteresis and time constants (engage within tens of milliseconds exploiting the electromechanical lead; release over 0.5-1 s to avoid chatter). Layer 2 (tremor estimator): band-power threshold to start; upgrade to WFLC/BMFLC on the EMG envelope or IMU signal to track the personal tremor frequency online. Layer 3 (impedance rendering): for semi-active hardware, map tremor power to damping/stiffness level (proportional semi-active control); for motorized hardware, impedance about the voluntary estimate from a two-stage (g-h plus WFLC) filter. Sampling 500 Hz-1 kHz for the EMG front end, 100-200 Hz control loop.

8.5 Safety and ergonomics requirements

From the literature: total hand-borne mass under 200-300 g (user studies place preference near 200 g; accepted ceiling 400-500 g); joint torque capability bounded (0.1 Nm-class active torque caps in comparable finger exoskeletons; our semi-active resistance should be sized to damp tremor torques, not to immobilize the finger against voluntary effort); hard end-stops inside anatomical range; open palm and fingertips to preserve touch; one-handed donning under 1-2 minutes for a tremulous user; skin pressure distributed over straps and shells; and instrumented monitoring for proximal tremor migration during trials [24, 46, 19].

Section 9 - Evaluation methodology

9.1 Signal-level metrics

Primary outcome: reduction of tremor band power (3.5-7.5 Hz) at the index fingertip/hand dorsum IMU, device-on versus device-off, in three conditions: rest, posture (outstretched holding), and grasp-and-hold (cup, spoon, pegs). Report percent power reduction per condition. Benchmarks: WOTAS 40% average (up to 80%); human-tested orthoses 70-86%;

WTSG 59-86% across joints and conditions; jamming glove 25-42% clinical. Secondary: tremor amplitude (RMS displacement, integrated from the IMU), and stability of suppression over time.

9.2 Detector metrics

Detection latency from burst onset (EMG ground-truthed manually or via a TKEO reference) to the stiffening command - target below the electromechanical delay window (under about 50-100 ms); sensitivity and false positives per minute during tremor-free voluntary tasks (reaching, typing, pouring); robustness across sessions (re-donning) and across the medication cycle if patients participate.

9.3 Task-level and human metrics

Standard hand-function tests with device on versus off: nine-hole peg test, spiral drawing (Archimedes spiral rating), cup-of-water transfer (spill mass), buttoning. Transparency test: interaction resistance during voluntary finger motion with the device worn but passive (subjective rating plus, if instrumented, interaction torque). Usability: donning time, comfort rating over 30-60 minute wear, weight perception. If patient access is not available in the project period, the accepted fallback used across this field is healthy subjects with simulated tremor (voluntary 4-6 Hz oscillation or recorded tremor replayed on a rig) - WTSG, HELM and the jamming glove were all first validated this way [34, 35, 32].

9.4 Ethics and practicalities

University ethics approval for EMG on healthy volunteers; patient testing only with clinical partners. sEMG is non-invasive and low-risk; the main practical constraints are skin prep, session length (fatigue), and data protection for physiological recordings.

Section 10 - Conclusion and recommended direction

The research supports this project direction: (1) target the grasp-and-hold scenario - re-emergent tremor makes static holding the unprotected phase of daily hand use; (2) act at the fingers - almost all existing devices stop at the wrist or the utensil, and finger-level suppression has been demonstrated only in a handful of research prototypes; (3) stiffen semi-actively rather than counter-actuate - comparable efficacy class (52-86% in the literature), inherent safety (cannot inject energy), lower mass and power; (4) trigger from EMG - it is the only sensing modality that precedes the mechanical tremor (20-100 ms electromechanical lead), it doubles as the grasp-intent sensor, and only one prior orthosis has used it at all; and (5) evaluate against the published benchmarks (40-86% band-power reduction) with both signal-level and task-level metrics. The combination - thumb+index, semi-active, EMG- and grasp-contingent - occupies a niche that no reviewed device fills, giving the project both engineering feasibility (every component class is proven separately) and genuine novelty (the combination is unexplored).

Section 11 - References

- [1] GBD 2021 Parkinson's disease burden. BMC Public Health, 2025.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12427116/>
- [2] Projections of PD prevalence to 2050. BMJ, 2025.
<https://pubmed.ncbi.nlm.nih.gov/40044233/>
- [3] Postuma RB et al. MDS clinical diagnostic criteria for Parkinson's disease. Movement Disorders, 2015. <https://movementdisorders.onlinelibrary.wiley.com/doi/10.1002/mds.26424>
- [4] Clinical neurophysiology of Parkinson's disease (tremor frequency bands). Clinical Neurophysiology Practice, 2022.
<https://www.sciencedirect.com/science/article/pii/S2467981X22000269>
- [5] Jankovic J et al. Re-emergent tremor of Parkinson's disease. JNNP, 1999; Mov Disord Clin Pract, 2016. <https://pmc.ncbi.nlm.nih.gov/articles/PMC1736624>
- [6] Hauser RA et al. Burden of tremor in Parkinson's disease: patient survey. J Parkinson's Disease, 2025. <https://pmc.ncbi.nlm.nih.gov/articles/PMC13347455/>
- [7] Accelerometer-based quantification of UPDRS tremor scores. IEEE EMBC, 2011.
<https://doi.org/10.1109/iembs.2011.6090126>
- [8] Tremor differential diagnosis reviews. American Family Physician, 2011/2018.
<https://www.aafp.org/pubs/afp/issues/2011/0315/p697.html>
- [9] Swinnen BEKS et al. Tremor is highly responsive to levodopa in advanced PD. Mov Disord Clin Pract, 2025.
<https://movementdisorders.onlinelibrary.wiley.com/doi/full/10.1002/mdc3.14262>
- [10] Zach H, Dirks MF et al. Dopamine-resistant tremor in Parkinson's disease. Neurology, 2020. <https://www.neurology.org/doi/10.1212/WNL.00000000000010316>
- [11] Levodopa-induced dyskinesia reviews, 2023.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10342913/>
- [12] Cury RG et al. Long-term outcomes of thalamic DBS for tremor. Neurology, 2017.
<https://www.neurology.org/doi/10.1212/WNL.0000000000004295>
- [13] Frey J et al. Global economic evaluation of DBS. Stereotact Funct Neurosurg, 2024.
<https://pubmed.ncbi.nlm.nih.gov/38513625/>
- [14] Bond AE et al. MRgFUS thalamotomy for tremor-dominant PD, randomized trial. JAMA Neurology, 2017. <https://pubmed.ncbi.nlm.nih.gov/29084313/>
- [15] Nguyen HS, Luu TP. Tremor-suppression orthoses for the upper limb. Frontiers in Human Neuroscience, 2021. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8119649/>
- [16] Pascual-Valdunciel A et al. Peripheral electrical stimulation to reduce pathological tremor: review. J NeuroEngineering and Rehabilitation, 2021.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7885254/>
- [17] Mo J, Priefer R. Medical devices for tremor suppression. Biosensors, 2021.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8065649/>
- [18] Rocon E, Pons JL et al. Design and validation of a rehabilitation robotic exoskeleton for tremor assessment and suppression (WOTAS). IEEE TNSRE, 2007.
<https://ieeexplore.ieee.org/document/4303108/>
- [19] Lora-Millan JS et al. Biomechanical loading as an alternative treatment for tremor. Tremor and Other Hyperkinetic Movements.
<https://tremorjournal.org/articles/10.5334/tohm.107>
- [20] Adabi K, Ondo W. Review of peripheral tremor devices. Tremor and Other Hyperkinetic Movements, 2024. <https://tremorjournal.org/articles/10.5334/tohm.930>

- [21] Pathak A et al. Noninvasive handheld device to cancel tremor (Liftware). Movement Disorders, 2014.
- [22] Brillman S et al. TAPS for PD action tremor. Tremor and Other Hyperkinetic Movements, 2023. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10453948/>
- [23] Gallego JA et al. Neuroprosthesis for tremor management through control of muscle co-contraction. JNER, 2013. <https://link.springer.com/article/10.1186/1743-0003-10-36>
- [24] Sarac M et al. Design requirements of generic hand exoskeletons. arXiv/IEEE ToH, 2019. <https://arxiv.org/html/1911.06408>
- [25] du Plessis T et al. A review of active hand exoskeletons. Robotics (MDPI), 2021. <https://www.mdpi.com/2218-6581/10/1/40>
- [26] Polygerinos P et al. Soft robotic glove for combined assistance and at-home rehabilitation. Robotics and Autonomous Systems, 2015.
- [27] In H et al. Exo-Glove: wearable robot for the hand with soft tendon routing. IEEE RAM, 2015.
- [28] Butzer T et al. Fully wearable hand exoskeleton (Tenoexo). Soft Robotics, 2021. <https://doi.org/10.1089/SORO.2019.0135>
- [29] Yurkewich A et al. HERO grip glove for stroke. JNER, 2020. <https://jneuroengrehab.biomedcentral.com/articles/10.1186/s12984-020-00659-5>
- [30] Zahedi A et al. Soft exoskeleton for tremor suppression with flexible semiactive actuator (SETS). Soft Robotics, 2021. <https://journals.sagepub.com/doi/abs/10.1089/soro.2019.0194>
- [31] Miniature MR brakes: WSU 0.9 Nm at 85 g; MDPI Actuators 0.3 Nm at 122 g. <https://vancouver.labs.wsu.edu/robotics-and-automation/miniature-mr-brake/>
- [32] Awantha WVI et al. Layer-jamming soft glove for hand tremor suppression. IEEE RA-L/ICRA, 2021. <https://pubmed.ncbi.nlm.nih.gov/34905493/>
- [33] Diller S, Majidi C, Collins SH. Lightweight low-power electroadhesive clutch. IEEE ICRA, 2016. <https://ieeexplore.ieee.org/document/7487194/>
- [34] Zhou Y, Naish MD, Trejos AL et al. Wearable tremor suppression glove (WTSG). IEEE TBME, 2021. <https://pubmed.ncbi.nlm.nih.gov/33999812/>
- [35] HELM: hand exoskeleton with linear motors for five-finger tremor suppression, 2025. <https://www.sciencedirect.com/science/article/pii/S3050544525000179>
- [36] Skaramagkas V et al. Pneumatic-muscle glove for index finger tremor attenuation, 2021.
- [37] Admittance/impedance control stability in physical human-robot interaction; backdrivable actuator surveys. <https://arxiv.org/pdf/1902.03966>
- [38] Mann KA et al. Frequency spectrum analysis of wrist motion for activities of daily living. J Orthopaedic Research, 1989. <https://pubmed.ncbi.nlm.nih.gov/2918428/>
- [39] Riviere CN, Thakor NV et al. Adaptive cancelling of physiological tremor (WFLC). IEEE TBME, 1998.
- [40] Veluvolu KC, Ang WT. Band-limited multiple Fourier linear combiner (BMFLC). IEEE EMBC, 2007. <https://ieeexplore.ieee.org/document/4352922/>
- [41] WAKE: wavelet decomposition with adaptive Kalman filtering for tremor extraction. arXiv, 2017. <https://arxiv.org/pdf/1711.06815>
- [42] Second-order-moment sEMG tremor onset detector (under 8 ms bias). Medical Engineering and Physics, 2012. <https://www.sciencedirect.com/science/article/abs/pii/S1350453311003444>
- [43] Free-living PD tremor detection with wrist wearables, 2024. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11129907/>

- [44] Widjaja F, Shee CY, Ang WT et al. Electromechanical delay for real-time anti-phase tremor attenuation using FES. IEEE ICORR, 2011.
<https://ieeexplore.ieee.org/document/5979865/>
- [45] Lobo-Prat J et al. Non-invasive control interfaces for intention detection in active movement-assistive devices. JNER, 2014. <https://pubmed.ncbi.nlm.nih.gov/25516421/>
- [46] Dickmann T et al. Adaptive mechatronic exoskeleton for force-controlled finger rehabilitation. Frontiers in Robotics and AI, 2021.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8514640/>
- [47] Dosen S et al. Online tremor suppression using EMG and low-level electrical stimulation. IEEE TNSRE, 2015.
- [48] De Luca CJ, Contessa P. Hierarchical control of motor units. J Neurophysiology, 2012.
<https://journals.physiology.org/doi/full/10.1152/jn.00961.2010>
- [49] De Luca CJ. Surface electromyography: detection and recording. Delsys tutorial, 2002; and Filtering the surface EMG signal, J Biomechanics, 2010.
<https://www.delsys.com/downloads/TUTORIAL/semg-detection-and-recording.pdf>
- [50] Leijnse JN et al. Anatomic basis for individuated surface EMG of the extensor digitorum communis. J Neurophysiology, 2008.
<https://journals.physiology.org/doi/full/10.1152/jn.00706.2007>
- [51] Non-invasive estimation of deep forearm muscle activity; intrinsic hand muscle recording practice, 2018.
- [52] Hermens HJ et al. SENIAM recommendations for surface electromyography. J Electromyogr Kinesiol, 2000. http://seniam.org/sensor_location.htm
- [53] De Luca CJ et al. Inter-electrode spacing: reduction of crosstalk. J Biomechanics, 2012.
<https://www.sciencedirect.com/science/article/abs/pii/S0021929011006920>
- [54] Hardware specifications: Delsys Trigno; MyoWare 2.0; BITalino; OYMotion gForce.
<https://myoware.com/products/technical-specifications/>
- [55] Smith LH, Hargrove LJ et al. Optimal window length for pattern-recognition myoelectric control. IEEE TNSRE, 2011.
- [56] Besomi M et al. CEDE consensus: EMG amplitude normalization. J Electromyogr Kinesiol, 2020. <https://pubmed.ncbi.nlm.nih.gov/32569878/>
- [57] Solnik S et al. TKEO improves EMG onset detection. Eur J Applied Physiology, 2010.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2945630/>
- [58] Tenore FVG et al. Decoding of individuated finger movements using surface EMG. IEEE TBME, 2009. <https://pubmed.ncbi.nlm.nih.gov/19473933/>
- [59] Al-Timemy AH et al. Classification of finger movements from forearm EMG. IEEE JBHI, 2013.
- [60] Shukla AW, Panyakaew P. Neurophysiological basis of tremor disorders. Seminars in Neurology, 2026. <https://pmc.ncbi.nlm.nih.gov/articles/PMC13297120/>
- [61] Nistico R et al. Synchronous pattern distinguishes resting tremor of ET from PD. Parkinsonism and Related Disorders, 2011. <https://pubmed.ncbi.nlm.nih.gov/21071257/>
- [62] Zhang J et al. Differential diagnosis of PD, ET and enhanced physiological tremor with EMG tremor analysis. Parkinson's Disease, 2017.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5573102/>
- [63] di Biase L et al. Tremor stability index. Brain, 2017.
<https://academic.oup.com/brain/article/140/7/1977/3782569>
- [64] Dideriksen JL et al. EMG-based characterization of pathological tremor using the iterated Hilbert transform. IEEE TBME, 2011. <https://pubmed.ncbi.nlm.nih.gov/21803679/>

- [65] Samaee S, Kobraei HR. Predicting the occurrence of wrist tremor from EMG. Biomedical Signal Processing and Control, 2019.
<https://www.sciencedirect.com/science/article/abs/pii/S1746809419303209>
- [66] Ruonala V et al. Levodopa-induced changes in EMG patterns in advanced PD. Frontiers in Neurology, 2018.
<https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2018.00035/full>
- [67] Young AJ, Hargrove LJ, Kuiken TA. Electrode shift sensitivity in myoelectric pattern recognition. IEEE TBME, 2011. <https://pubmed.ncbi.nlm.nih.gov/21659017/>